What Metric Scale Can a Frozen Foundation-Model Depth Prior Give Monocular SLAM?

Abstract— **Problem. Monocular SLAM has no absolute scale of its own. A frozen FM metric-depth network looks like it could supply one, and is increasingly used as if it does. We ask the narrow, falsifiable question: *does it actually transfer metric scale, and if not, what is the smallest fix?***

What we found. On HI-SLAM2 with MapAnything over ten sequences (sixteen for the gate), it does not, and the reason is structural. The global depth rescale a metric prior would supply is an exact blind spot of the monocular factor graph (3.4×10^{-13} px reprojection change), so no reprojection, photometric, or prior-consistency residual (and no online estimator) can observe it. HI-SLAM2 realizes this gauge freedom by median-normalizing the prior at every keyframe and hard-coding the initial gauge; removing those mechanisms (a true metric-depth factor, Sec. V-G) leaves the conclusion unchanged, which is what makes the diagnosis a property of the monocular graph rather than of one codebase. No cheap image signal replaces an anchor: neither cross-view depth consistency nor a prior-free depth ratio predicts the needed correction, and an online estimator beats a fixed model on one of three sequences. Two constructive results survive. A trust gate on the *dispersion* of a short prior-free warm-up ratio flags when the network's scale is untrustworthy (AUROC 0.97 on sixteen sequences; the network's own confidence scores 0.44). And a trajectory-scale anchor recovers scale on all ten validated sequences (mean error 0.024) and thirteen of sixteen overall, where a depth-mean anchor fails (mean error 0.306). A trust-gated protocol combining the two cuts mean trajectory error from 5.9 m to 0.11 m on the ten (14.7 m to 0.33 m across all sixteen).

What to do. Treat the metric prior as a sensor to be *audited*, not trusted: gate its scale on the warm-up dispersion, and if it is trusted, anchor the gauge to the model. Otherwise spend a sparse, one-shot external metric measurement (an occasional rangefinder, stereo frame, or fiducial check) to pin trajectory scale, and use the model for dense structure only. The identifiability diagnosis and the trajectory anchor generalize across depth models; the dispersion gate is specific to MapAnything, and GT-free detection of scale uncertainty remains open.

I. Introduction

Monocular SLAM reconstructs a scene from a single moving camera. Because a photograph fixes no absolute size, the whole reconstruction is only defined up to one global factor; we call this the gauge. A reconstruction can be perfectly self-consistent and still be $3\times$ too small. Something external must fix the gauge before the output is metric.

Foundation-model monocular depth networks are an appealing external source. Many of them, including MapAnything and the metric variants of Depth Anything, predict depth in metres, and are now dropped into SLAM and Gaussian-splatting pipelines as if those metres transfer through the pipeline unchanged. This paper asks whether they do.

a) Names used throughout.: The frozen depth network is the *teacher* (MapAnything; DA-V2 Metric-Indoor and DA3METRIC for generality). The SLAM system it feeds is the *vehicle* (HI-SLAM2). The outside measurement that pins the global size is the *anchor*; we compare a *depth-mean* anchor (match the scene's average depth) with a *trajectory-scale* anchor (match a metric translation to the trajectory). r_{init} is the median ratio of teacher depth to the system's own depth over a short prior-free warm-up, and *dispersion* is its spread; AUROC scores the trust gate (0.5 = chance); ATE is trajectory error in metres after rigid alignment with scale fixed, so a scale error appears as trajectory error.

b) Setting and use case.: The camera that runs continuously is monocular, but in many robotics deployments a metric measurement is available *occasionally*: a downward rangefinder or imaging sonar [1], a stereo or RGB-D frame that is briefly mounted or momentarily reliable [2], a fiducial at a docking station [3], or GNSS in good sky [4]. Fusing episodic measurements of this kind is the classical intermittent-observation setting [5]. We take that literally: the protocol below spends roughly twenty sparse keyframe pairs *once* at startup to fix the gauge, then tracks with the monocular camera and the frozen FM prior for dense geometry. This is not continuous multi-sensor fusion; it is a bounded, one-shot metric check placed exactly where monocular SLAM is structurally blind.

c) What we did, and the short answer.: We study one concrete SLAM system (HI-SLAM2) and one frozen depth model (MapAnything) over ten sequences (sixteen for the gate), then repeat the key test with two more depth models (DA-V2 Metric-Indoor, DA3METRIC) to see what generalizes; KITTI is a documented out-of-domain control, reported as a limit of the SLAM system rather than a result about depth models. The metric scale does not transfer, and the reason is not that one system is buggy: it is an *identifiability* problem. The SLAM system realizes the monocular gauge freedom by normalizing the prior at every keyframe and fixing the initial gauge to a constant, so the model's metres are discarded before they can matter; we show the same holds after those mechanisms are removed (Sec. V-G). You must supply one external metric number. Once you do, the pipeline is metric to the accuracy of that number. The practical version is a trust gate plus a trajectory-scale anchor.

d) What we found (in order of the paper).: (i) The system discards the prior's scale by construction: swapping depth networks changes nothing (0.316 vs. 0.339), because

a hard-coded gauge constant dominates. (ii) Anchoring the gauge to the model recovers scale in-domain to the model's own bias (0.907 vs. 0.903) but transfers the wrong scale out of distribution (EuRoC 0.13–0.21). (iii) No free signal replaces the anchor, and an online estimator beats a fixed model on 1 of 3 sequences. (iv) The *spread* of the warm-up ratio is a usable trust gate *for MapAnything* (AUROC 0.97 over sixteen sequences; the model's own confidence scores 0.44). (v) The right anchor is trajectory scale, not depth mean: the latter fails (0.70/0.86/0.86 on three EuRoC sequences), the former recovers all ten (mean error 0.024) and thirteen of sixteen, failing only on difficult low-parallax sequences. (vi) A deliberate 25% local FM-depth step is absorbed by both the stock and the metric-factor vehicle: there is no live local lever. (vii) Gate + anchor cut mean trajectory error 5.9 m $\rightarrow$ 0.11 m; the diagnosis and anchor hold across three models, the specific gate does not.

e) Contributions.: (1) A measured identifiability result: the global metric-scale direction is an exact null direction of the monocular factor graph, so no reprojection, photometric, or prior-consistency residual (nor any online estimator) can recover it without an absolute metric measurement (Sec. III-B; confirmed after removing the vehicle's keyframe normalization, Sec. V-G). (2) A universal fallback: a trajectory-scale anchor that recovers scale on thirteen of sixteen sequences, where a depth-mean anchor fails (Sec. V-E). (3) A demonstrator and a fence: model-internal confidence is consistently useless for scale trust (AUROC 0.44; a constant 1.0 for DA3METRIC), while the dispersion of the initial ratio is a viable gate *for MapAnything* (AUROC 0.97); GT-free scale-uncertainty detection remains open and model-dependent (Secs. V-C–V-D).

II. Related Work

We position this as a *measured identifiability result and a protocol*, not a new SLAM back-end. The FM-prior SLAM line (HI-SLAM2 [6], CF2-SLAM [7], PRISM-SLAM [8], Scalix [9], VGGT-SLAM 2.0 [10]) reaches metric monocular output by *adopting* the teacher's scale, directly or through a calibrated probabilistic factor, and works when the teacher is in-domain. None can detect when adoption fails: PRISM's ray-distance factor makes the metric scale Fisher-identifiable but assumes the prior carries the right metres; Scalix treats a *learned* per-frame scale uncertainty as an independent measurement; CF2-SLAM calibrates coverage, not the scale value. Our audit supplies the missing failure statement, our **trust gate** the missing failure detector, and our **anchor** the missing fallback.

The similarity gauge and monocular scale unobservability are classical; we give the concrete consequence for FM depth priors, a numeric certificate, and its low-parallax face. The visual-inertial line [11], [12] estimates learned-depth scale in *initialization* because the IMU makes scale observable, exactly the external observable that pure monocular SLAM lacks and that our anchor supplies. We study metric-depth models as geometry/Gaussian priors: MapAnything [13] is our frozen teacher, with DA-V2 Metric [14] and DA3METRIC [15] for generality. We do not claim to introduce metric SLAM, learned covariance [16], or scale-aware factor graphs, and this is not a bug report about HI-SLAM2: its normalization lines instantiate a gauge freedom that any anchor-free monocular pipeline has, as confirmed after they are removed (Sec. V-G).

III. How the System Handles Depth, and Why Scale is a Blind Spot

A. The setup

HI-SLAM2 is monocular factor-graph SLAM with a dense depth prior fed to a depth-only bundle adjustment at a weight of 0.01. The depth prior comes from a frozen map of depth images (one per frame) produced by the teacher, in the system's working resolution.

B. The gauge argument, and a numeric check

We state the fact we rely on once, as a proposition with a one-paragraph proof, because the corollary (not the proposition) is what the paper uses. The similarity gauge is classical; nothing below is claimed as new mathematics.

Proposition 1 (gauge null direction). *Take a pixel with depth z_i, intrinsics K, and relative pose (R, t). The reprojection residual of its backprojected point is invariant to the joint rescaling $(t, z) \rightarrow \lambda(t, z)$ for every $\lambda > 0$.*

Proof sketch. Backproject $X_i = z_i K^{-1}[u_i, 1]$ and project forward as $X_j = RX_i + t$. Under the joint rescale, $X_j \rightarrow \lambda X_j$, and projection divides out that common scale, so the image point u_j does not move, while the depth it implies grows by λ. Only the ratio t/z enters the residual. Depth-only rescaling (z alone, t fixed) is not a symmetry, because it changes that ratio. □

Corollary 1. *Any estimator state whose update direction is the joint rescale (a global depth-scale "bias" state) lies in the exact null space of the residual and is structurally unobservable. In pure monocular SLAM such a state carries no information and should not be estimated; the global metric scale can be set only by an external metric reference.* This corollary is the paper's diagnosis: it is *why* the trust gate needs a prior-free reference and *why* an anchor is mandatory, and it bounds what any online estimator can certify. Corollary 1 is a statement about the *monocular factor graph*, not about a particular implementation: reprojection, photometric, and prior-consistency residuals are all invariant to the joint (t, z) rescale, so no arrangement of them, including a metric-depth factor at full weight, observes the scale direction. A graph with no absolute metric measurement retains this gauge freedom, and an estimate initialized in that gauge stays there.

We checked the proposition numerically. A joint (t, z) rescale changes the reprojected point by 3.4×10^{-13} px (numerical zero). A depth-only rescale (z alone, t fixed) changes it by 215.6 px, and a small rotation perturbation by 6.8 px, so the test is sensitive. The three numbers together are the certificate: the joint direction is null, the depth-only direction is not, and the measurement could have seen a difference if there were one.

C. How the shipped vehicle realizes the gauge freedom

Two mechanisms are load-bearing. First, at every keyframe the prior is median-normalized to the system's own disparity (the ratio of the two medians). Second, at warm-up the mean depth is pinned once to a constant (a scale multiplier of 2.0). Together these mean the teacher's absolute scale never reaches the estimate, and the global gauge is set once and frozen. These lines are *how this vehicle realizes* the gauge freedom, not the result: Sec. V-G removes them by wiring the prior as a true metric-depth factor, and the global conclusion is unchanged, exactly as the structural argument predicts.

IV. Method

1) **Audit.** Serve a frozen teacher and confirm the realized scale does not depend on which teacher it is (we swap Omnidata for MapAnything and get the same result).
2) **Gauge anchor.** Replace the hard-coded gauge constant with the teacher's mean metric depth. The measured relation is $\mathrm{scale}(a) = \mathrm{scale}(1)/a$, so the oracle anchor is $a = \mathrm{scale}(1)$. In-domain this makes the reconstruction metric to the teacher's own accuracy.
3) **Trust gate.** From a short *prior-free* warm-up, compute $r_{init} = \mathrm{median}(D_{teacher}/D_{backend})$ and its dispersion over the window. Trust the teacher's scale when the dispersion is low.
4) **Sparse anchor protocol (opportunistic sensor fusion).** When the gate rejects a sequence (or it is known to be out of distribution), pin metric **trajectory scale** from one short external metric translation (RGB-D or stereo PnP, matched to the backend trajectory) and use the teacher for dense structure only. A depth-mean anchor is not sufficient.

V. Experiments

a) Setup.: HI-SLAM2 (torch 2.1.2) plus frozen MapAnything depth maps over ten sequences: TUM f1_xyz, f1_desk, fr2_xyz, fr2_desk, fr3_long_office_household; EuRoC MH_01, MH_02, V1_01, V1_02, V2_01. All injection is external (no edits to the SLAM code); ground truth is TUM registered depth and TUM/EuRoC trajectories.

b) Vehicle-domain control.: KITTI-800 (odometry 00, first 800 frames, outdoor driving) is a documented limit of the *SLAM system*, not a result about depth models: stock HI-SLAM2 gives umeyama scale 4.74 (ATE 10.3 m) and the FM-wired run 3.86 (ATE 13.9 m). Both are wrong by roughly $4\times$ regardless of the prior, so we exclude KITTI from the prior claims.

A. The gauge dominates; anchoring recovers scale in-domain

Table I shows the effect. On TUM, anchoring recovers the teacher's own accuracy (0.907 vs. 0.903): the dominant error was the hard-coded gauge, not the network. On EuRoC the teacher's scale is itself wrong, so the same anchor transfers the wrong metres. The gauge law is linear to measurement precision: anchoring f1_xyz at 1.1 gives $0.907/1.1 = 0.825$ (measured 0.825) and f1_desk at 0.8 gives $1.063/0.8 = 1.327$ (measured 1.327). Read the adopted scale against the teacher-bias column, not against 1.0: the residual is the teacher's bias faithfully inherited.

TABLE I

Umeyama scale (est→GT) before and after anchoring the gauge to the teacher, and the teacher's own average bias; "fixed gauge" pins mean depth to 2.0 m.

sequence	fixed gauge	anchored	teacher bias
f1_xyz	0.442	**0.907**	0.903
f1_desk	0.505	**1.063**	0.907
MH_01	0.316	**0.132**	out of distribution
V1_02	0.192	**0.210**	out of distribution

TABLE II

Scale error of the five-arm ladder. Online adaptation beats frozen calibration on only one of three informative sequences.

sequence	fixed gauge	in-domain	frozen	online	oracle
f1_xyz	0.496	0.093	0.107	0.177	0.000
f1_desk	0.403	0.063	0.047	0.130	0.000
MH_01	0.661	0.868	0.870	0.682	0.000
V1_02	0.808	0.790	0.794	0.977	0.000

B. No free signal replaces the anchor

The candidate was the warm-up ratio $r_{init} = \mathrm{median}(D_{teacher}/D_{backend})$ from a prior-free run. Its *average* does not predict the correction: the pairwise sign agrees three times and disagrees three times, and a fit calibrated on one domain extrapolates $18\times$ wrong on MH_01. A five-arm ladder (fixed gauge, in-domain, frozen calibration, online adaptation, oracle) agrees in scale error (Table II): the online arm beats the frozen arm on only one of three informative sequences. **We rejected the online-transfer route.**

C. Dispersion is a usable trust gate (for MapAnything)

The *spread* of r_{init} does work. With an out-of-distribution label defined purely from metric truth ($|\mathrm{scale} - 1| > 0.30$), low dispersion means "trust" and scores **AUROC 0.917** on ten sequences (threshold 0.0625, accuracy 0.90), with one conservative false rejection. The label is not the domain: V1_01 and V2_01 are EuRoC yet trustworthy, and fr2_xyz is TUM yet out of distribution. The network's own confidence is useless (AUROC 0.44): it is confidently wrong. A sequence-level bootstrap gives 95% CI [0.667, 1.000], leave-one-out never drops below 0.889, and a fixed frame budget leaves the AUROC at 0.917 (frame-count correlation -0.26): the gate reads domain shift, not sample size.

Adding the six remaining EuRoC sequences (**16 total, 7 gate-positive**) raises the AUROC to **0.968** (95% CI [0.859, 1.000]; leave-one-out ≥ 0.963; fixed budget 0.968). The added sequences are informative: MH_03/04/05 are flagged out of distribution (dispersion 0.51/0.64/0.80) and V1_03/V2_02/V2_03 trusted (0.09/0.06/0.02). The added reference runs

TABLE III

GATE AUROC PER DEPTH MODEL (TEN SEQUENCES). BOOTSTRAP CIS OVERLAP BECAUSE THERE ARE ONLY 4–6 OOD CASES; LEAVE-ONE-OUT SEPARATES THEM.

depth model	AUROC	95% CI	LOO
MapAnything	**0.917**	[0.667, 1.000]	0.889–1.000
DA-V2 Metric-Indoor	0.625	[0.167, 1.000]	0.500–0.778
DA3METRIC-LARGE	0.600	[0.160, 1.000]	0.500–0.700

TABLE IV

ANCHOR ERROR PER SEQUENCE FOR A DEPTH-MEAN VERSUS A TRAJECTORY-SCALE ANCHOR, WITH THE TEACHER'S OWN ERROR.

sequence	sensor (m)	teacher	depth-mean	traj.
f1_xyz	1.128	0.093	0.113	**0.009**
f1_desk	1.163	0.063	0.020	**0.069**
fr2_xyz	1.330	0.359	0.029	**0.044**
fr3_long	2.532	0.005	0.106	**0.010**
MH_01	2.275	0.868	0.704	**0.004**
MH_02	3.248	0.903	0.856	**0.047**
V1_02	2.696	0.790	0.857	**0.013**
V2_01	3.414	0.203	0.156	**0.031**
mean (10)		0.358	0.306	**0.024**

use a GS-free, 300-frame vehicle, so this compares the init-window statistic across a slightly different backend.

D. The gate is model-specific

Two more metric depth models confirm the identifiability diagnosis and fail differently. **DA-V2 Metric-Indoor-Large** [14] is less metric (6 of 10 out of distribution, including two TUM); **DA3METRIC-LARGE** [15] is near-metric in-domain but over-estimates by 15–43% (5 of 10 OOD) and is badly off on EuRoC (0.23–0.26). DA3METRIC even emits a constant confidence of 1.0, so confidence is never a scale-trust signal.

Table III reports the comparison. Leave-one-out is the sharper statistic: both alternative models fall to chance (0.50) while MapAnything never drops below 0.889. Allowing each model its own threshold does not close the gap: MapAnything reaches AUROC 1.00 at a stable $\tau^* = 0.20$ (LOSO 0.215 ± 0.045), while DA-V2 and DA3METRIC reach only 0.875 ($\tau^* = 0.475$) and 0.810 ($\tau^* = 0.700$) at unstable τ^* (LOSO std 0.144/0.135). **The identifiability diagnosis and the anchor generalize; the dispersion gate is MapAnything-specific.**

E. The right anchor is trajectory scale, not depth mean

With the gauge law, an oracle anchor gives scale error $\epsilon/(1+\epsilon)$ on every sequence, independent of the scene. A realistic *depth-mean* anchor (matching average scene depth) does not (Table IV). Measured with real backend runs using each dataset's own sensor:

The depth-mean anchor is no better than the teacher overall (0.306 vs 0.358) and catastrophic on three EuRoC sequences (0.70/0.86/0.86), because the average depth that makes the sensor metric and that makes the trajectory metric are different. A trajectory-scale anchor (a metric translation matched to the backend trajectory) recovers all ten sequences (mean **0.024**), using 11–27 matched pairs with a bootstrap interval. Extended to the six remaining EuRoC sequences it accepts **13 of 16** (mean 0.021, worst 0.069), refusing MH_04, MH_05 and V2_03, the hardest sequences with the poorest stereo pairs. The anchor is exact where matched translation exists and refuses where it does not; the teacher-only failure is a fact about identifiability, not low parallax (a 10% depth-scale error changes reprojection by 2.3×10^{-13} px at zero translation and grows monotonically with baseline).

F. The sparse-anchor protocol, and what it costs

Composing the gate with the trajectory anchor gives the practical result, measured as rigid-aligned trajectory error (ATE) so that a scale error shows up as metric error. **Teacher scale only** (the naive default of trusting an FM prior's metres) gives mean scale error 0.358 and mean ATE **5.94 m**; **gate + trajectory anchor** gives 0.060 and **0.11 m**. This is *catastrophe prevention*: on the four OOD sequences teacher-only produces tens-of-metres errors (MH_01 $30.0 \rightarrow 0.05$ m, MH_02 $21.4 \rightarrow 0.11$ m, V1_02 $6.7 \rightarrow 0.07$ m). Figs. 1 and 2 summarize the gate and protocol. Across all sixteen sequences the protocol cuts mean ATE from **14.7 m to 0.33 m** (scale 0.417→0.083); the residual is entirely the difficult cases where the anchor fails (MH_04 $66.7 \rightarrow 2.0$ m, MH_05 $59.5 \rightarrow 1.5$ m). The protocol degrades boundedly rather than silently.

The gate needs only a short prior-free warm-up and the anchor only 11–27 sparse keyframe pairs, applied *once* at startup; thereafter tracking is pure monocular with the FM prior supplying dense geometry. The metric source can be opportunistic (a rangefinder or sonar, a brief stereo or RGB-D frame, a docking fiducial, GNSS) and is not committed to the front end [5], [1], [2], [3], [4]. The one case this does not help is a platform whose only sensor is RGB-D.

G. The global gauge has no local lever

If the global gauge is a blind spot, perhaps local model scale is trackable. It is not, in either vehicle. In the shipped system a deliberate 25% step in the model's depth from frame 400 leaves the global scale unmoved (0.8246 vs. 0.8252) and the per-keyframe depth pass-through at the noise floor (∼0.0016), because of the per-keyframe renormalization. Wired as a true **metric-depth factor** (scale grid removed, prior entering at its own metres), the global result is unchanged: the scalar and confidence-weighted arms sit at or slightly worse than stock (mean $|\text{scale}-1| = 0.523$ vs. 0.409) and never approach the anchored result; raising the prior weight α from 0.01 to 10 reproduces the $0.44 \rightarrow 0.50$ shift but never reaches the teacher's 0.90 (Table V). **The failure survives removing the per-keyframe normalization entirely, so it is the gauge null of the monocular graph, not HI-SLAM2's implementation.** A deliberate local step is likewise absorbed by both vehicles, and per-pixel confidence weighting changes shape, not scale.

H. The observable half, and a designed blind spot

A scalar variance scale on the observable half can be calibrated online: weighting the metric-factor residual by

TABLE V

UMEYAMA SCALE (EST→GT) PER METRIC-FACTOR ARM. THE METRIC RESIDUAL DOES NOT MAKE THE GLOBAL GAUGE OBSERVABLE: THE SCALAR AND CONFIDENCE-WEIGHTED ARMS NEVER APPROACH THE ANCHORED RESULT.

sequence	stock	scalar	conf.	metric+anch.	anchor only
f1_xyz	0.505	0.505	0.499	0.896	0.907
f1_desk	0.597	0.577	0.538	0.964	1.063
fr2_xyz	0.680	0.748	0.750	0.708	0.641
fr2_desk	1.000	1.088	1.090	0.823	0.808
fr3_long	1.132	1.450	1.432	1.288	1.005
MH_01	0.339	0.270	0.353	0.133	0.132
MH_02	0.231	0.271	0.265	0.125	0.097
V1_01	1.065	1.333	1.351	1.403	1.107
V1_02	0.192	0.212	0.242	0.036	0.210
V2_01	1.439	1.881	2.008	1.607	1.203
mean	0.409	0.517	0.523	0.462	0.358

TABLE VI

SCALAR VERSUS K=3 PER-REGIME ADAPTATION: DEPTH-CHANNEL MEDIAN NIS (TARGET 1) AND PER-REGIME NIS. PER-REGIME IS CLOSER TO 1 BUT NOT ROBUST TO K.

sequence	scalar med.	regime med.	scalar NIS_k	regime NIS_k
f1_xyz	1.380	1.315	[0.60,2.00,2.78]	[1.33,1.34,1.24]
f1_desk	0.469	0.707	[0.12,0.37,1.05]	[0.50,0.36,0.67]
MH_01	0.033	0.504	[0.01,0.01,0.04]	[0.04,0.07,0.67]
V1_02	0.025	0.615	[0.01,0.01,0.03]	[0.16,0.17,0.64]

TABLE VII

THE GATE IS BLIND TO A CONSTANT TEACHER BIAS: PERTURBING THE INTRINSICS SHIFTS THE ADOPTED SCALE BUT BARELY MOVES THE GATE STATISTIC.

		dispersion		adopted scale	
sequence	gate	real	pert.	real	pert.
f1_xyz	trust	0.012	0.019	0.907	0.877
f1_desk	trust	0.022	0.026	1.063	1.022
MH_01	reject	0.240	0.260	0.132	0.120

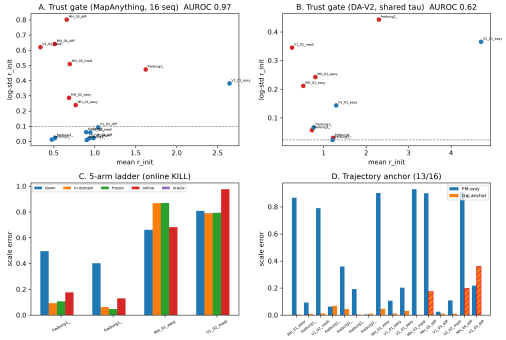


Fig. 1. (A) trust gate, MapAnything, sixteen sequences (AUROC 0.97); (B) trust gate, DA-V2 Metric at a shared threshold (AUROC 0.62; the gate is model-specific); (C) the five-arm ladder (the online route we rejected); (D) trajectory anchor versus teacher-only across sixteen sequences; the hatched bars are the three sequences the anchor rejects.

the teacher's confidence and adapting a variance scale by covariance matching brings the depth-channel median NIS closer to 1 than frozen weighting on every sequence (e.g. MH_01 0.021 → 0.033, V1_02 0.082 → 0.025), though it pins its floor on extreme-OOD MH_01. A per-regime (K=3) extension passed its pre-registered test in one configuration but is **not robust** (Table VI): unguarded K2/K4 hung in the backend and a stability guard erased the gain, so we report only the scalar adaptation. The calibration is also domain-specific, and depth-channel NIS is circular for absolute scale by construction.

Finally, the gate is **blind to a constant teacher bias**: perturbing the intrinsics ($fx, fy \times 1.20$) shifts the adopted scale (0.907→0.877) while dispersion barely moves (Table VII), because r_{init} is a ratio against the backend and a constant bias rescales it uniformly. This is a designed limitation: the gate detects teacher inconsistency/OOD, and a constant bias must be caught by the anchor. Fig. 3 collects these five supporting results.

VI. DISCUSSION AND LIMITATIONS

a) Why the free-signal route fails.: The reference used by r_{init} is itself only up-to-scale, so the relation between r and the needed correction requires a domain-independent reference gauge that monocular geometry does not provide. This is the observability statement, not a bug in our implementation.

b) Two halves of the error; scale is not adaptive.: Corollary 1 splits the prior's error into a part the residuals can see (per-pixel σ, the relative scale across keyframes, the shape) and a part they cannot: the single global scale direction. A scalar part of the observable half is a legitimate target for online estimation and consistency checking (Sec. V-H); the unobservable half cannot be estimated at all and must instead be *gated* and *anchored* by an external reference. This is why the two constructive pieces do not overlap, and why the honest framing for the global direction is one-shot identifiability plus an anchor, not adaptive recovery of scale. Depth-channel consistency checks are *circular* for absolute scale (the prior defines the scale they would validate), and image-channel residuals are blind to the joint rescale (Proposition 1): an adaptive estimator can certify only what the gauge lets it see.

c) Practical protocol.: (1) Gate the model's metric scale on the dispersion of the prior-free warm-up ratio. (2) If trusted, anchor the gauge to the model. (3) Otherwise pin trajectory scale with a metric translation and use the model for dense structure only; a depth-mean anchor is not sufficient.

d) What generalizes, and what does not.: The gauge diagnosis (Corollary 1) and the trajectory anchor hold across the three depth models and both in- and out-of-distribution domains; these are the transferable results. The trust gate does not: the dispersion of the initial ratio separates scale errors for MapAnything (AUROC 0.97) but not for DA-V2 (0.62) or DA3METRIC (0.60), and model-internal confidence is useless everywhere we looked (0.44, or a constant 1.0). General GT-free detection of metric-scale uncertainty is therefore an open problem, not a solved one.

e) Limitations.: Sixteen sequences (five TUM, eleven EuRoC), three metric models, one SLAM system. The six

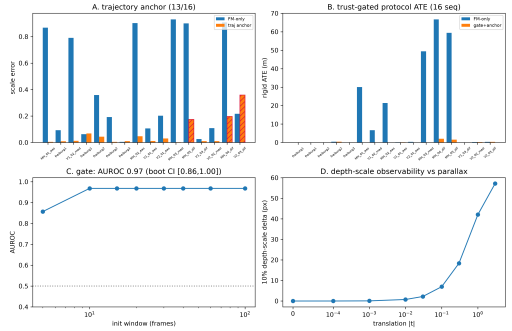


Fig. 2. (A) trajectory anchor versus teacher-only, sixteen sequences (13 accepted); (B) trust-gated protocol error across sixteen sequences, teacher-only versus gate+anchor; (C) gate init-window sweep with bootstrap interval (sixteen sequences); (D) depth-scale visibility versus parallax.

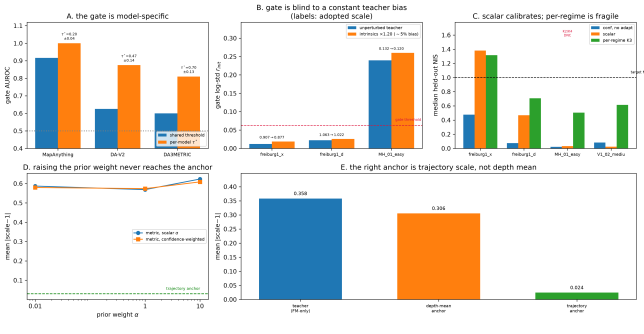


Fig. 3. The five results stated above, shown together: (A) the gate is model-specific: a per-model threshold lifts DA-V2/DA3METRIC to 0.88/0.81 but at unstable τ^* (LOSO std 0.14/0.13), while MapAnything reaches 1.00 at a stable $\tau^* = 0.20$; (B) the gate is blind to a constant teacher bias; (C) scalar adaptation reaches NIS ≈ 1 while the per-regime extension is fragile (K2/K4 did not converge unguarded); (D) raising the prior weight α through three orders of magnitude leaves the metric-factor arms near mean $|\text{scale} - 1| \approx 0.6$, far from the trajectory anchor's 0.03; (E) of the three anchors only the trajectory one works: teacher-only 0.358, depth-mean 0.306, trajectory 0.024 mean $|\text{scale} - 1|$ over ten sequences.

added EuRoC reference runs use a GS-free, 300-frame vehicle configuration (the Gaussian backend crashes on some of those visuals), so that part of the gate comparison is across a slightly different backend setup. The gate is model-specific and must be validated per model; it detects teacher inconsistency, not a constant bias, which by construction leaves the gate statistic unchanged (Sec. V-H), so a globally biased teacher must be caught by the anchor. Local-drift testing is on TUM only (EuRoC has no dense ground-truth depth); the trajectory anchor still needs an external metric sensor, and it recovers 13 of the 16 sequences, refusing the three difficult EuRoC sequences where matched translation is too poor to certify a scale; the sixteen-sequence gate and protocol figures include those failures rather than excluding them. KITTI is reported as a limit of the SLAM system, not a result about depth models. The per-regime σ of Sec. V-H passes the pre-registered test in one configuration but is **not robust**: K2/K4 hung unguarded and a stability guard erased the calibration gain, so only the scalar adaptation is reported as a mechanism. A deliberate local FM step is likewise absorbed by the metric-factor vehicle (Sec. V-G), so we report no live local lever.

VII. Conclusion

A frozen FM metric-depth prior does not transfer metric scale to monocular SLAM on its own. The monocular gauge makes the *global* scale direction unobservable (Corollary 1), and the system as shipped normalizes away even local model scale per keyframe. The diagnosis is implementation-independent: it is confirmed after those normalizations are removed (Sec. V-G), so it is a property of the monocular factor graph, not of one codebase. For the global direction the fix is one external metric anchor plus a GT-free trust gate, and no cleverer bias or covariance estimator can substitute. This is a statement about scale, not about adaptation in general: a scalar variance scale on the observable half can be calibrated online, though a per-regime refinement was not robust, provided neither is mistaken for scale recovery. We give the measured chain and a minimal protocol.

References

[1] D. Yang, H. Ai, J. Liu, and B. He, "Absolute scale estimation for underwater monocular visual odometry based on 2-d imaging sonar," *Measurement*, vol. 190, p. 110665, 2022.

[2] R. Mur-Artal and J. D. Tardós, "Orb-slam2: An open-source slam system for monocular, stereo, and rgb-d cameras," *IEEE Trans. Robotics*, vol. 33, no. 5, pp. 1255–1262, 2017.

[3] E. Olson, "Apriltag: A robust and flexible visual fiducial system," in *IEEE Int. Conf. Robotics and Automation (ICRA)*, 2011, pp. 3400–3407.

[4] K. J. Wu, C. X. Guo, G. Georgiou, and S. I. Roumeliotis, "Vins on wheels," in *IEEE Int. Conf. Robotics and Automation (ICRA)*, 2017, pp. 5155–5162.

[5] B. Sinopoli, L. Schenato, M. Franceschetti, K. Poolla, S. S. Sastry, and M. I. Jordan, "Kalman filtering with intermittent observations," *IEEE Trans. Automatic Control*, vol. 49, no. 9, pp. 1453–1464, 2004.

[6] W. Zhang, Q. Cheng, D. Skuddis, N. Zeller, D. Cremers, and N. Haala, "Hi-slam2: Geometry-aware gaussian slam for fast monocular scene reconstruction," *IEEE Trans. Robotics*, 2025.

[7] X. Chen, "Cf2-slam: Conformal-calibrated foundation-factor graph slam across modalities and domains," *Computers, Materials & Continua*, 2026.

[8] E. Im, G. Lee, S. Hong, and J. Suh, "Prism-slam: Probabilistic ray-grounded inference for scale-aware metric slam," *arXiv preprint arXiv:2605.19257*, 2026.

[9] S. B. Laina, T. Zhang, P. Petropoulakis, S. Schaefer, S. Boche, J. Jung, C. L. Gentil, and S. Leutenegger, "Scalix: Uncertainty-aware scale-consistent monocular slam," *arXiv preprint arXiv:2608.17553*, 2026.

[10] D. Maggio and L. Carlone, "Vggt-slam 2.0: Real-time dense feed-forward scene reconstruction," *arXiv preprint arXiv:2601.19887*, 2026.

[11] N. Merrill, P. Geneva, S. Katragadda, C. Chen, and G. Huang, "Fast and robust learned single-view depth-aided monocular visual-inertial initialization," *Int. J. Robotics Research*, vol. 44, no. 10-11, pp. 1619–1647, 2025.

[12] H. Hailu, B. Gebregziabher, S. Gudipudi, and M. Bhatta, "Quantifying motion excitation for metric scale observability in monocular visual-inertial odometry," *arXiv preprint arXiv:2603.26740*, 2026.

[13] N. Keetha, N. Müller, J. Schönberger, L. Porzi, Y. Zhang, T. Fischer, A. Knapitsch, D. Zauss, E. Weber, N. Antunes, J. Luiten, M. Lopez-Antequera, S. R. Bulò, C. Richardt, D. Ramanan, S. Scherer, and P. Kontschieder, "Mapanything: Universal feed-forward metric 3d reconstruction," *arXiv preprint arXiv:2509.13414*, 2025.

[14] L. Yang, B. Kang, Z. Huang, X. Wang, J. Feng, and H. Zhao, "Depth anything v2," *arXiv preprint arXiv:2406.09414*, 2024.

[15] L. Yang, B. Kang, Z. Huang, X. Zhao, K. Zhao, H. Zhao, and et al., "Depth anything 3," *arXiv preprint arXiv:2504.11310*, 2025.

[16] Y. Qiu, Y. Chen, Z. Zhang, W. Wang, and S. Scherer, "Mac-vo: Metrics-aware covariance for learning-based stereo visual odometry," *arXiv preprint arXiv:2409.09479*, 2024.